

Geometric and hemodynamic determinants of functional tricuspid regurgitation: A real-time three-dimensional echocardiography study

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Abstract

Background: The geometric and hemodynamic determinants of functional tricuspid regurgitation (FTR) have not yet been established. Geometric alterations in the tricuspid valve can be accurately determined using 3-dimensional echocardiography.

Methods: Real-time 3-dimensional echocardiography was performed in 54 patients with various degrees of FTR to obtain real-time zoom 3D images of the tricuspid valve. The angles between the tricuspid annulus plane and the 3 leaflets (anterior: α , posterior: β , septal: γ), the septal–lateral and anterior–posterior annulus diameters, and the tricuspid tenting volume were measured on a mid-systole frame. Pulmonary artery systolic pressure (PASP) was assessed using the maximal velocity of FTR, and the severity of FTR was quantified as the ratio of the distal jet area to the right atrial area (DJ/RA) using color Doppler 2D echocardiographic images.

Results: In patients with mild ($\text{DJ/RA} < 0.2$) and moderate ($0.2 \leq \text{DJ/RA} < 0.5$) FTR, β and γ were significantly larger than α , whereas in patients with severe ($\text{DJ/RA} \geq 0.5$) FTR, only γ was larger than α . All geometric variables and PASP correlated significantly with DJ/RA. Multiple stepwise regression analysis showed that γ , septal–lateral annulus diameter and PASP were independent determinants of FTR severity. The tricuspid tenting volume strongly correlated ($r=0.84$, $p<0.001$) with the FTR distal jet area, and was the only determinant of the jet area.

Conclusion: FTR severity is mainly determined by septal leaflet tethering, septal–lateral annular dilatation, and the severity of pulmonary hypertension.

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Keywords: Tricuspid valve regurgitation; Severity; Three-dimensional echocardiography

1. Introduction

Functional tricuspid regurgitation (FTR), which is often accompanied by various left-side valvular heart disease and myocardial disease has been reported to have an ominous prognosis and to limit exercise capacity [1–6]. Suggested FTR pathogenesis includes geometric alterations of the tricuspid valve apparatus, such as tricuspid annulus dilation and valvular tenting [7–14]. Hemodynamic factors such as pulmonary artery systolic pressure (PASP) were also reported to be associated with FTR severity [1,2,15–17].

Most previous studies have used 2-dimensional echocardiography for analyzing tricuspid valve geometry. However, this approach has some limitations since the 2-dimensional image is subject to the location of the imaging plane which in turn is dependent on the examiner as well as the patient's chest window. In addition, it is sometimes difficult to identify which portion of which leaflet is being observed in the 2-dimensional imaging plane. Such limitations can result in a subjective and inaccurate analysis of tricuspid valvular geometry. In contrast, more precise and objective imaging planes crossing the 3 leaflets of the tricuspid valve and annulus can be acquired using real-time 3-dimensional echocardiography (RT3DE). However, tricuspid valve geometric analysis in FTR patients using RT3DE remains to be demonstrated.

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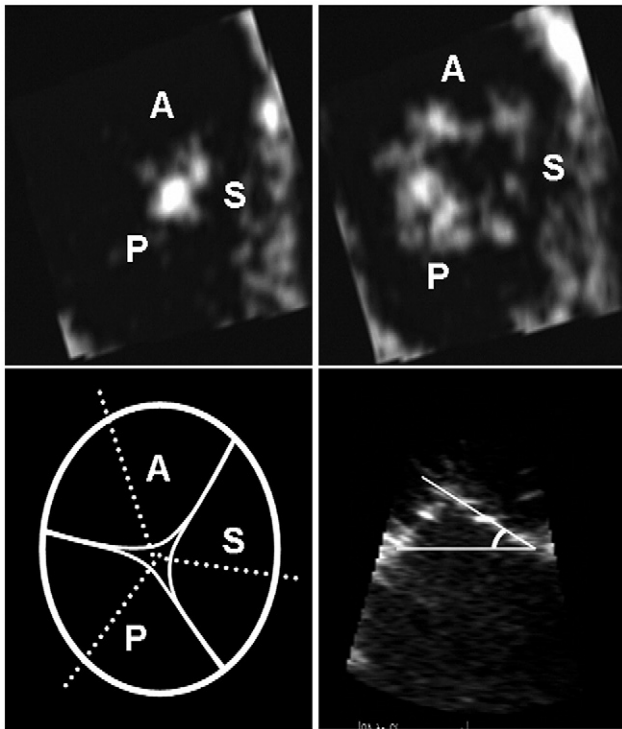


Fig. 1. A representative example of measuring angles between the tricuspid annulus plane and the 3 leaflets using a real-time 3-dimensional zoom image. In a mid-systole frame, the 3 tricuspid valve leaflets were identified on the short axis plane at the level of the valve tip (upper left panel) and mid-valve level (upper right panel). Three longitudinal planes (dotted lines, lower left panel) which perpendicularly crossed the middle of each leaflet were generated to measure the angles between the tricuspid annulus line and each leaflet (lower right panel). A, P and S represent anterior, posterior and septal leaflets, respectively.

Although significant FTR is usually surgically managed using tricuspid annuloplasty, such surgery does not always successfully eliminate FTR [8,18–20]. It is likely that more accurate geometric and hemodynamic investigation into FTR can help to solve this difficult clinical problem. Therefore, we sought to evaluate the main geometric and hemodynamic determinants of FTR using RT3DE and Doppler echocardiography.

2. Materials and methods

A total of 54 patients (27 females, age: 60 ± 13 years) with various degrees of FTR were enrolled in the study. Mitral valve disease and left ventricular dysfunction were present in 16 and 10 patients, respectively. The remaining patients had no significant left-side heart disease. Thirty-nine of the 54 patients showed chronic atrial fibrillation, while 15 presented with a normal sinus rhythm. Patients with poor echocardiographic images or thickened and restricted tricuspid valves were excluded from this study.

RT3DE was performed to obtain a real-time zoom 3-dimensional image of the tricuspid valve from the low parasternal or apical view using a SONOS 7500 (Philips,

Andover, MA). Care was taken to include the tricuspid annulus and valve leaflets in the zoom image set. Real-time zoom 3-dimensional imaging was used as it could be acquired during 1 cardiac cycle and did not require 4 regular consecutive cycles to form. Thus, this 3-dimensional imaging tool could be applied to patients with atrial fibrillation. In patients with atrial fibrillation, 3 images were acquired during the 3 different beats. The RT3DE images were digitally stored on the hard disk in the echocardiography machine, and transferred to a personal computer for off-line analysis.

Using dedicated software (Tomtec, Munich, Germany), 3 longitudinal planes which perpendicularly crossed the middle of each leaflet were generated with guidance of the short axis image of the tricuspid valve (Fig. 1). On each of the 3 longitudinal planes, the angles between the tricuspid annulus line and the 3 leaflets (anterior: $A\alpha$, posterior: $P\alpha$, septal: $S\alpha$) and the lengths of the 3 leaflets were measured on a mid-systole frame. The septal–lateral and antero-posterior longitudinal planes which were perpendicular to each other were created to measure septal–lateral and antero-posterior tricuspid annulus diameters (Fig. 2). The tricuspid tenting volume was also measured at mid-systole by tracing tenting

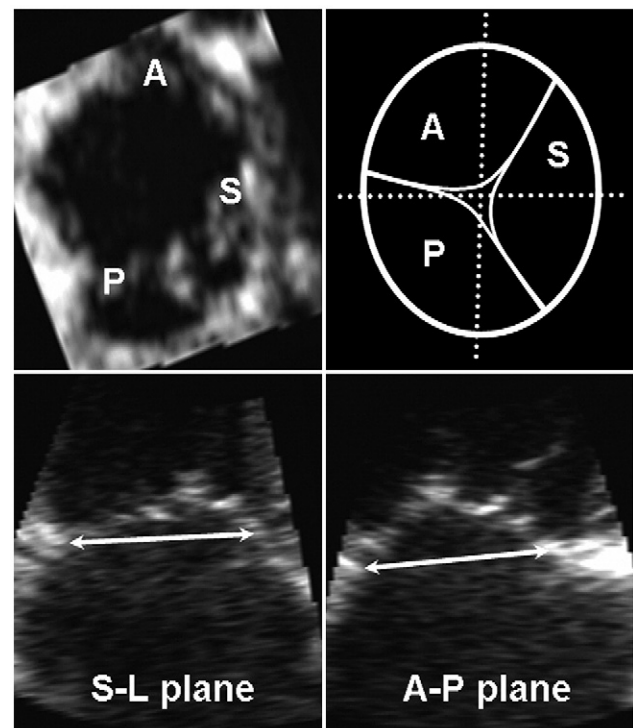


Fig. 2. A representative example of measuring the septal–lateral (S–L) and antero-posterior (A–P) tricuspid annulus diameters using a real-time 3-dimensional zoom image. In a mid-systole frame, the tricuspid annulus shape and orientation were assessed using the short axis image of the annulus (upper left panel). The S–L and A–P longitudinal planes (dotted lines, upper right panel) which were perpendicular to each other were created to measure the S–L (lower left panel) and A–P (lower right panel) tricuspid annulus diameters. A, P and S represent anterior, posterior and septal leaflets, respectively.

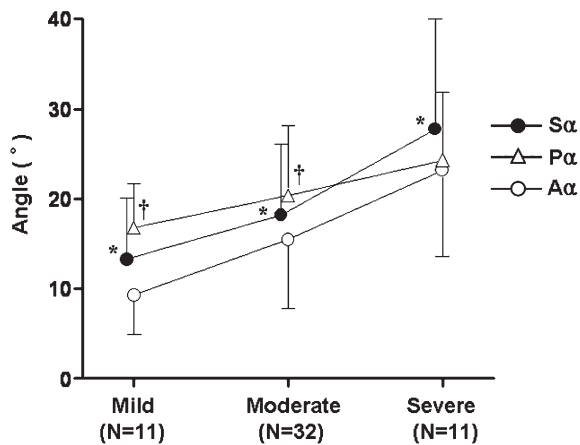


Fig. 3. The angles between the tricuspid annulus plane and the 3 leaflets (anterior: Aα, posterior: Pα, septal: Sα) at mid-systole in patients with mild, moderate and severe functional tricuspid regurgitation. * and † denote $p < 0.05$ and $p < 0.005$ vs. the corresponding Aα, respectively.

areas in 8 radial planes 22.5° apart from one another [21]. In patients with atrial fibrillation, 3 measurements in 3 different beats were averaged.

Using 2-dimensional Doppler echocardiography, continuous-wave Doppler tracing of FTR was performed and the maximal velocity of FTR was determined to assess the PASP using a modified Bernoulli equation [22]. Right atrial pressure was estimated as 5 mm Hg if the inferior vena cava (IVC) was not dilated (<1.7 cm) and there was a 50%

Table 1

Univariate and multivariate regression analyses for determinants of the functional tricuspid regurgitation severity based on the ratio of the distal jet area to the right atrial area

	<i>r</i>	Univariate <i>P</i>	Multivariate <i>P</i>
Aα	0.48	<0.001	0.30
Pα	0.31	<0.05	0.58
Sα	0.46	<0.001	<0.05
Septal–lateral annulus diameter	0.47	<0.001	<0.05
Antero–posterior annulus diameter	0.46	<0.001	0.58
Tricuspid tenting volume	0.49	<0.001	0.30
Pulmonary artery systolic pressure	0.48	<0.001	<0.05

Aα: the angle between the tricuspid annulus plane and anterior leaflet.

Pα: the angle between the tricuspid annulus plane and posterior leaflet.

Sα: the angle between the tricuspid annulus plane and septal leaflet.

decrease in the diameter during inspiration, 10 mm Hg if the IVC was dilated with normal inspiratory collapse, and 15 mm Hg if the IVC was dilated and did not collapse with inspiration [23]. On a mid-systole frame, the FTR distal jet area and the right atrial area were measured using color Doppler images, and the FTR severity was quantified based on the ratio of the distal jet area to the corresponding right atrial area (DJ/RA) [2,8]. Mild, moderate and severe FTR were defined as a DJ/RA <0.2 , $0.2 \leq \text{DJ/RA} < 0.5$ and DJ/RA ≥ 0.5 , respectively. In patients with atrial fibrillation, 3 measurements in 3 different beats were averaged.

Statistical analysis was performed using SPSS 12.0 (SPSS Inc, Chicago, IL). Data are expressed as mean $\pm$ SD.

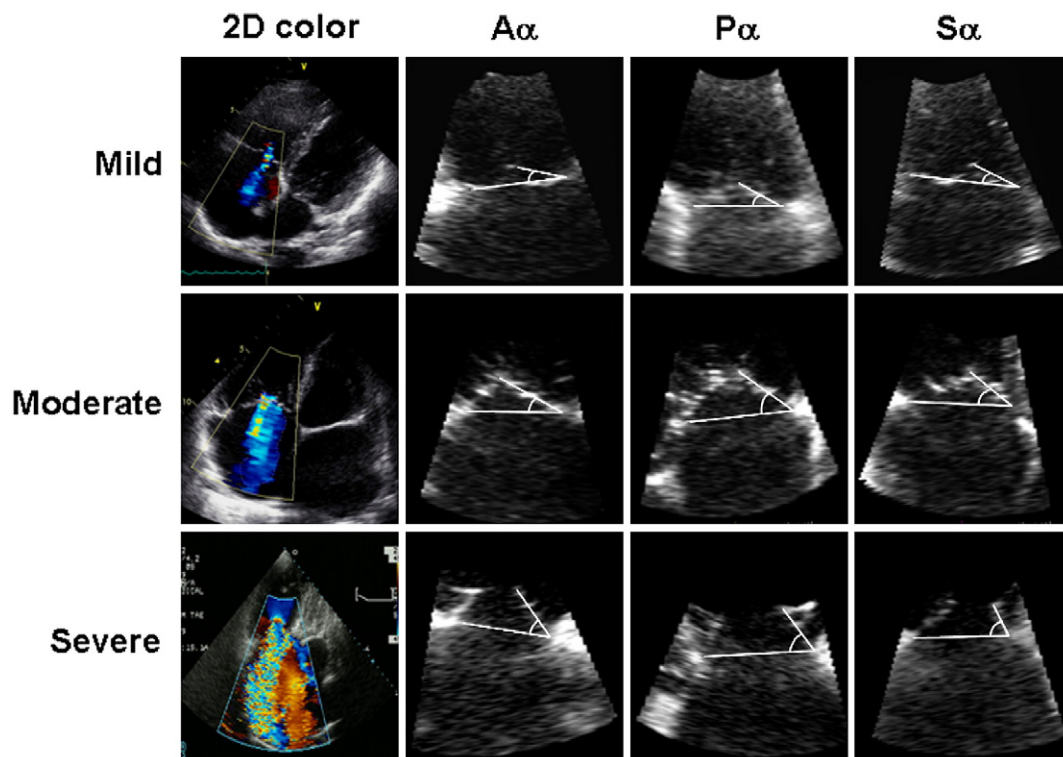


Fig. 4. Representative examples of color Doppler images and the angles between the tricuspid annulus plane and the 3 leaflets (anterior: Aα, posterior: Pα, septal: Sα) at mid-systole in patients with mild, moderate and severe functional tricuspid regurgitation.

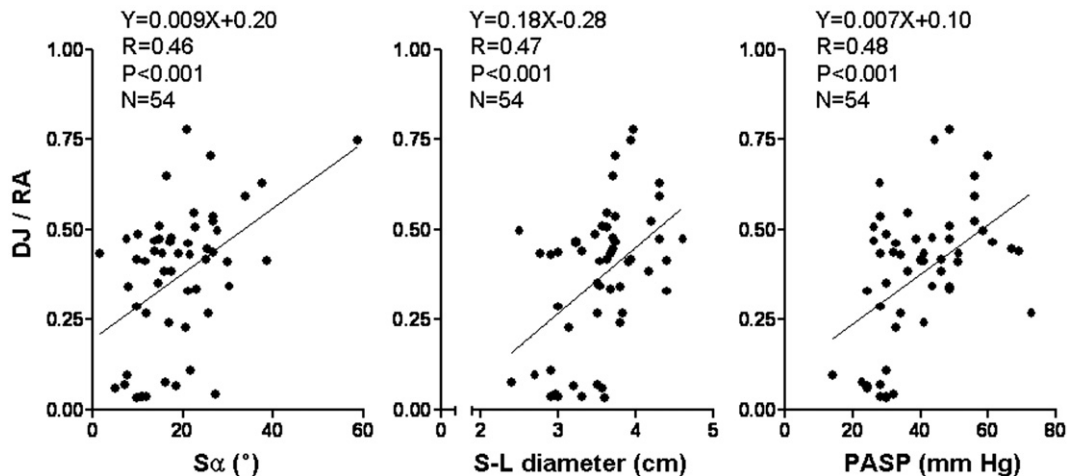


Fig. 5. Correlations of the angle between the tricuspid annulus plane and the septal leaflet ($S\alpha$), septal–lateral (S–L) annulus diameter and pulmonary artery systolic pressure (PASP) with the severity of functional tricuspid regurgitation quantified based on the ratio of the distal jet area to the right atrial area (DJ/RA).

Comparisons between 3 variables were performed using repeated-measures ANOVA. Comparisons between 2 measurements were evaluated using paired *t* tests. We assessed univariate regression with DJ/RA or distal jet area of FTR as a dependent variable and other measurements. Then, multivariate analysis based on stepwise multiple regression analysis was implemented to assess independent determinants among measurements with significant univariate correlations ($P < 0.05$). A *P* value < 0.05 was considered significant.

3. Results

In all 54 patients, anterior leaflet and posterior leaflet lengths were significantly larger than septal leaflet lengths (2.1 ± 0.3 , 2.0 ± 0.4 vs. 1.8 ± 0.3 cm, respectively; $p < 0.001$). $P\alpha$ and $S\alpha$ were larger than $A\alpha$ (20.4 ± 7.5 , 19.1 ± 9.8 vs. $15.8 \pm 8.7^\circ$, respectively; $p < 0.001$). In 11 patients with mild FTR, $P\alpha$ ($16.8 \pm 4.9^\circ$, $p < 0.005$) and $S\alpha$ ($13.3 \pm 6.8^\circ$, $p < 0.05$) were significantly larger than $A\alpha$ ($9.3 \pm 4.4^\circ$) (Figs. 3 and 4). Similarly, in 32 patients with moderate FTR, $P\alpha$ ($20.4 \pm 7.8^\circ$, $p < 0.001$) and $S\alpha$ ($18.2 \pm 7.9^\circ$, $p < 0.05$) were larger than $A\alpha$ ($15.5 \pm 7.7^\circ$). In 11 patients with severe FTR, only $S\alpha$ ($27.8 \pm 12.2^\circ$, $p < 0.05$) was significantly

larger than $A\alpha$ ($23.2 \pm 9.6^\circ$), while $P\alpha$ ($24.3 \pm 7.5^\circ$, $p = 0.32$) was not.

$A\alpha$, $P\alpha$, $S\alpha$, septal–lateral and antero-posterior annulus diameters, tricuspid tenting volume and PASP were found to correlate with DJ/RA (Table 1). Multiple stepwise regression analysis showed that $S\alpha$, septal–lateral annulus diameter and PASP were independent determinants of FTR severity (Fig. 5). In 15 patients with sinus rhythm, the septal–lateral annulus diameter was the only determinant of FTR ($p < 0.005$), while $S\alpha$ ($p < 0.005$) and PASP ($p < 0.05$) were determinants in patients with atrial fibrillation. In mild FTR patients, those with atrial fibrillation ($n = 4$) showed significantly larger septal–lateral annulus diameter (3.4 ± 0.3 vs. 2.9 ± 0.3 cm, $p < 0.05$) and larger antero-posterior diameter (3.5 ± 0.5 vs. 2.9 ± 0.4 cm, $p = 0.06$) with marginal significance than those with normal sinus rhythm ($n = 7$). In moderate-to-severe FTR patients, there was no significant difference between the two groups in septal–lateral (3.6 ± 0.4 vs. 3.9 ± 0.6 cm, $p = 0.26$) and antero-posterior (3.7 ± 0.5 vs. 3.7 ± 0.9 cm, $p = 0.99$) annulus diameters.

$A\alpha$, $P\alpha$, $S\alpha$, septal–lateral and antero-posterior annulus diameters, tricuspid tenting volume and PASP showed significant correlations with the FTR distal jet area (Table 2). Multiple stepwise regression analysis showed that the tricuspid tenting volume was the only independent determinant of the FTR distal jet area. $S\alpha$ was the only determinant ($p < 0.05$) in the 15 patients with sinus rhythm, while the tenting volume was the only determinant in those with atrial fibrillation ($p < 0.001$).

4. Discussion

FTR is usually managed surgically using various methods of tricuspid annuloplasty. However, FTR has been reported to persist post-operatively in up to 45% of patients, and even progress during the long-term follow-up period after surgery [18–20]. We investigated the determinants of FTR using

Table 2
Univariate and multivariate regression analyses for determinants of the functional tricuspid regurgitation distal jet area

	<i>r</i>	Univariate <i>P</i>	Multivariate <i>P</i>
$A\alpha$	0.59	< 0.001	0.37
$P\alpha$	0.41	< 0.005	0.08
$S\alpha$	0.68	< 0.001	0.34
Septal–lateral annulus diameter	0.39	< 0.005	0.81
Antero-posterior annulus diameter	0.65	< 0.001	0.40
Tricuspid tenting volume	0.84	< 0.001	< 0.001
Pulmonary artery systolic pressure	0.33	< 0.05	0.91

$A\alpha$: the angle between the tricuspid annulus plane and anterior leaflet.
 $P\alpha$: the angle between the tricuspid annulus plane and posterior leaflet.
 $S\alpha$: the angle between the tricuspid annulus plane and septal leaflet.

RT3DE and Doppler echocardiography to find a clue to solve this clinical issue.

Tricuspid tethering/tenting has been regarded as an important geometric cause of FTR [8,11,13]. The present study demonstrated that the degree of tenting correlated with FTR severity in all 3 leaflets. More importantly, the study found that septal leaflet tenting tended to be the most prominent among the 3 leaflets in patients with severe FTR, and that the degree of septal leaflet tenting was an independent determinant of FTR severity. The reason of this phenomenon is still speculative. The septal leaflet might be more easily tethered by geometric change and dilatation of the right ventricle. A possible cause might be that the size of the septal leaflet is relatively small compared with the anterior and posterior leaflets, as confirmed in this study. Since the 3 leaflet tentings related to the FTR severity, and septal leaflet tenting was a major determinant of FTR, an adjuvant surgical procedure to relieve leaflet tethering, especially of the septal leaflet, may be necessary in addition to annuloplasty in patients with severe FTR and a high risk of remnant FTR after surgery [8].

Previous reports found an association between tricuspid annulus dilation and the development and severity of FTR [10–14]. Some investigators considered severe tricuspid annulus dilation rather than FTR severity as an indicator for tricuspid annuloplasty [7,9]. The current study demonstrated that the septal–lateral and antero–posterior annulus diameters correlated with FTR severity, and that the septal–lateral annulus diameter was an independent determinant of FTR. This result is consistent with a recent RT3DE study demonstrating that the tricuspid annulus was more circular due to dilation along the free wall aspect in patients with significant FTR than in patients without FTR [24]. Based upon the present data, the aim of tricuspid annuloplasty should be focused on decreasing the septal–lateral annulus diameter (i.e. reducing the free wall side dilatation), which can also enhance the coaptation between the severely tethered septal leaflet and the other 2 leaflets.

Atrial fibrillation appears to be a cause and aggravating factor of FTR, especially after left-side valve surgery [25–28]. Atrial fibrillation is frequently associated with significant FTR. While a previous RT3DE study evaluated FTR in patients with sinus rhythm only [24], the present study also analyzed the tricuspid valve and annulus geometry in patients with atrial fibrillation using RT3DE zoom images. In addition, the present study also measured tenting angles of the three leaflets, which were not evaluated in the previous study. Atrial fibrillation and FTR were reportedly linked by annulus dilatation [25]. However, the present study found that the septal–lateral annulus diameter was the only determinant in patients with sinus rhythm, while $S\alpha$ and PASP were both determinants in those with atrial fibrillation. We also found that in mild FTR patients, those with atrial fibrillation had a significantly larger septal–lateral annulus diameter compared to those with a normal sinus rhythm. These results suggest that mild annulus dilatation due to atrial fibrillation usually does

not cause significant FTR, and additional tricuspid tethering, higher PASP or more severe annulus dilation appears to be required to develop significant FTR in patients with atrial fibrillation. This finding in FTR is accordant with a previous report regarding functional mitral regurgitation and atrial fibrillation [29].

Pulmonary hypertension was reported to be associated with FTR [17]. FTR attenuates after reducing pulmonary hypertension by pulmonary thromboembolectomy or percutaneous mitral valvuloplasty [2,15,16]. The present study showed that the severity of pulmonary hypertension is a determinant of FTR severity as well as other geometric variables.

We firstly demonstrated the relationship between the tricuspid tenting volume and FTR using RT3DE. Tenting volume correlated with FTR severity and was the only determinant for the FTR distal jet area. The tricuspid tenting volume appears to be a good and comprehensive geometric parameter for assessing the severity of morphological alterations in FTR since it reflects annulus dilatation and tenting of the 3 leaflets.

The present study had some limitations. Analysis of geometric deformation of the right ventricle and its relationship with FTR were not determined as it was not possible to acquire a full volume image of the right ventricle in patients with atrial fibrillation. Analysis of right ventricular geometric alterations may provide better insight into the etiology of tricuspid valvular geometric change, and provide more practical clinical implications of tricuspid repair in FTR patients.

A small study population with subdivision analyses is another limitation of this study. The present study did not involve a control group. This decision was based on the desire to analyze FTR severity and geometric parameters as continuous variables and to evaluate their correlations rather than undertake a dichotomous analysis. In general, a trace-to-mild degree of FTR, which was included in this study, is considered to be a normal range.

Although the study found statistically significant correlations between FTR severity and certain variables, these correlations were not particularly strong. This may have been due to issues involving the FTR severity quantification method using 2-dimensional color Doppler images and the quite complex structures of right ventricular papillary muscles and the tricuspid valve apparatus [30]. Since these structures have substantial individual variations, the geometric analysis of tricuspid valve would be better to be individualized especially prior to surgery for FTR. We believe that RT3DE is a useful tool for objective and accurate geometric analysis of FTR in this clinical situation.

In conclusion, this RT3DE study suggests that FTR severity is mainly determined by septal leaflet tethering, septal–lateral annular dilatation and the severity of pulmonary hypertension. This finding may lead to procedures that might improve the success rate of tricuspid repair in FTR patients.

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